

SIMATS ENGINEERING

ASSESSMENT TOOL 2: INDUSTRY PROBLEM-SOLVING TASK

Course Code: CSA51 | Course Name: Cryptography and Network Security

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Assessment Weightage: 20% (Total Marks: 25)

Code of Conduct Declaration:

I, T.Lakshmi Prasad Naidu (Reg No: 192324187), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: T.Lakshmi Prasad Naidu

1. SSL/TLS Overhead and Web Server Performance Optimization

Problem Context & Given Parameters

- Web Server Protocol: TLS 1.3
- HTTP Response Payload: 2,000 Bytes
- TLS Record Overhead (Record Header + Auth Tag): 25 Bytes
- TLS Handshake Overhead: 3 KB per session (3,072 Bytes [binary] / 3,000 Bytes [decimal])
- Request Traffic Volume: 10,000 HTTP requests per second
- New TLS Session Handshakes: 1,000 handshakes per second

Step-by-Step Mathematical Evaluation

A. Total Data Transmitted per Second (including TLS Overhead):

1. Total size per encrypted HTTP response record = Payload + Record Overhead
= 2,000 Bytes + 25 Bytes = 2,025 Bytes/response
2. Data transmission rate for HTTP responses = 10,000 req/s × 2,025 Bytes
= 20,250,000 Bytes/sec (20.25 MB/s)
3. Handshake transmission rate = 1,000 handshakes/s × 3,072 Bytes
= 3,072,000 Bytes/sec (3.072 MB/s)
4. Total Data Transmitted = 20,250,000 + 3,072,000 = 23,322,000 Bytes/sec
= 23.322 MB/s ($\approx$ 186.58 Mbps)

B. Percentage Overhead Due to TLS Encryption:

1. Baseline raw payload data rate = 10,000 req/s × 2,000 Bytes = 20,000,000 Bytes/sec (20.00 MB/s)
2. Total TLS Overhead (Record + Handshake) = 23,322,000 - 20,000,000 = 3,322,000 Bytes/sec
3. Total TLS Overhead Percentage = $(3,322,000 / 20,000,000) \times 100 = 16.61\%$
(Note: The record-layer framing overhead alone is $25 / 2000 = 1.25\%$; handshakes contribute the remaining 15.36%).

C. Bandwidth Consumed Due to Handshake Overhead per Second:

- Handshake Bandwidth = 1,000 handshakes/s × 3,072 Bytes = 3,072,000 Bytes/sec
- Handshake Bandwidth in Mbps = $(3,072,000 \times 8) / 10^6 = 24.576$ Mbps
- Handshake overhead represents 13.17% of total aggregate network bandwidth.

D. Security vs. Latency Trade-Off Analysis & Proposed Optimizations:

- Trade-Off Dynamics: TLS provides confidentiality, integrity, and server authentication at the expense of asymmetric cryptographic CPU compute during key exchange (ECDHE) and round-trip network delays (1-RTT in TLS 1.3 vs 2-RTT in TLS 1.2).
- Optimization 1: TLS 1.3 0-RTT Session Resumption (Pre-Shared Keys - PSK): Returning clients transmit early HTTP request data in the first flight, eliminating connection handshake latency.
- Optimization 2: HTTP/2 & HTTP/3 (QUIC) Multiplexing: Consolidates multiple bidirectional HTTP streams across a single connection, drastically lowering the handshake rate from 1,000/s to single-digit reconnects.
- Optimization 3: OCSP Stapling & Hardware Offloading: Server caches CA certificate revocation status (OCSP) to eliminate external client lookup latency, while Intel QAT / AES-NI offloads cryptographic processing from primary CPU cores.

2. PGP-Based Email Encryption Load in Enterprise System

Problem Context & Given Parameters

- Enterprise Email Volume: 5,000 emails per day
- Email Attachment Size: 10 MB per email ($10 \times 1,048,576 = 10,485,760$ Bytes binary / 10,000,000 Bytes decimal)
- PGP Cryptographic Overhead: 800 Bytes per email (Session Key Encryption + Digital Signatures)

Step-by-Step Mathematical Evaluation

A. Total Data Transmitted per Day (including Overhead):

1. Total size per encrypted email = Attachment Size + PGP Overhead
= 10,485,760 Bytes + 800 Bytes = 10,486,560 Bytes (10.00076 MB)
2. Total Data per Day = 5,000 emails × 10,486,560 Bytes
= 52,432,800,000 Bytes = 52.433 GB (or 48.832 GiB / 50.004 GB decimal)

B. Percentage Increase in Storage Due to PGP Overhead:

1. Baseline unencrypted daily attachment storage = $5,000 \times 10 \text{ MB} = 50,000 \text{ MB}$
2. Daily PGP overhead storage = $5,000 \times 800 \text{ Bytes} = 4,000,000 \text{ Bytes} = 3.815 \text{ MB}$
3. Percentage Storage Increase = $(800 \text{ Bytes} / 10,485,760 \text{ Bytes}) \times 100 = 0.00763\%$ ($\approx 0.008\%$)
(Practical Insight: If Base64 ASCII Armor formatting is enforced, transmission payload expands by $\sim 33.3\%$; raw binary OpenPGP packets maintain the minimal 0.008% overhead).

C. Computational Overhead Analysis during Encryption/Decryption:

- Hybrid Architecture Burden: PGP performs symmetric bulk encryption (AES-256) on the 10 MB stream and asymmetric operations (RSA-2048/4096 or Ed25519) to sign the hash digest and encrypt the symmetric session key.
- Decryption Bottleneck: Decryption requires RSA modular exponentiation with the recipient

private key, consuming heavy CPU clock cycles per message compared to stream symmetric decryption.

- Memory & I/O Overhead: Buffering 10 MB attachments in memory for DEFLATE compression, SHA-256 hashing, and cipher block chaining creates high memory pressure during peak outbound bursts.

D. Recommended Enterprise Optimization Strategies:

- Pre-Encryption Compression (DEFLATE / LZMA): PGP standard applies lossless compression before encryption. For text, PDF, and office documents, this yields 30–60% file size reduction, far outweighing the 800-byte security overhead.
- Asynchronous Queue Batching: Decouple mail submission (SMTP) from cryptographic rendering using distributed task queues (e.g., Celery / Redis / RabbitMQ worker pools) to smooth out CPU spikes.
- Cryptographic Hardware Acceleration (HSM / AES-NI): Utilize dedicated PCIe Hardware Security Modules (HSMs) for private key operations and leverage CPU AES-NI instruction sets for symmetric bulk data processing.

3. IPSec Transport vs Tunnel Mode Performance Comparison

Problem Context & Given Parameters

- Original IP Packet: 1,000 Bytes payload + 20-Byte IP Header = 1,020 Bytes
- Transport Mode Overhead: 30 Bytes (ESP Header, IV, Trailer, ICV/Auth Tag)
- Tunnel Mode Overhead: 50 Bytes (New Outer 20-Byte IP Header + 30-Byte ESP Overhead)
- Network Traffic Volume: 20,000 packets per second (pps)

Step-by-Step Mathematical Evaluation

A. Total Packet Size for Both Modes:

1. Original IP Packet Size = 1,000 Bytes (Payload) + 20 Bytes (Original IP Header) = 1,020 Bytes
2. Transport Mode Packet Size = Original Packet (1,020 Bytes) + Transport ESP Overhead (30 Bytes)
= 1,050 Bytes
3. Tunnel Mode Packet Size = Original Packet (1,020 Bytes) + Tunnel ESP & Outer Header Overhead (50 Bytes)
= 1,070 Bytes

B. Bandwidth Usage for Each Mode per Second:

1. Transport Mode Bandwidth:
 - In Bytes/sec = 20,000 pps × 1,050 Bytes = 21,000,000 Bytes/sec (21.00 MB/s)
 - In Mbps = (21,000,000 × 8) / 10⁶ = 168.00 Mbps
2. Tunnel Mode Bandwidth:
 - In Bytes/sec = 20,000 pps × 1,070 Bytes = 21,400,000 Bytes/sec (21.40 MB/s)
 - In Mbps = (21,400,000 × 8) / 10⁶ = 171.20 Mbps
3. Bandwidth Differential: Tunnel mode consumes an additional 400,000 Bytes/s (3.20 Mbps / 1.90% increase).

C. Comparative Efficiency and Security Matrix

Metric / Parameter	IPSec Transport Mode	IPSec Tunnel Mode
Encapsulation	Encrypts only the transport payload (L4); original IP header remains exposed.	Encrypts the entire original IP packet (L3+L4) inside an entirely new Outer IP datagram.
Throughput / MTU	Higher throughput (168 Mbps); lower risk of MTU fragmentation (1050 B vs 1500 B standard).	Slightly higher bandwidth (171.2 Mbps); requires MSS clamping to avoid fragmentation.
Traffic Analysis	Low Protection: Eavesdroppers inspect source/destination host addresses directly.	High Protection: Hides internal host IPs and network topology behind VPN gateways.
NAT Compatibility	Prone to NAT traversal failure without explicit NAT-T UDP encapsulation.	Native compatibility with gateway-to-gateway subnet routing and NAT topologies.

D. Deployment Recommendations & Justifications:

- Internal Corporate Communication: Recommend Transport Mode. Since traffic resides within trusted LAN boundaries/private VLANs, hiding internal IP headers is redundant. Transport mode minimizes header overhead, saves 3.20 Mbps bandwidth, and eliminates gateway routing encapsulation delays.
- External / WAN Communication: Recommend Tunnel Mode. Across the public Internet or untrusted inter-site links, Tunnel Mode is mandatory. It encapsulates private RFC 1918 subnets, protects internal topology against reconnaissance/traffic analysis attacks, and enables secure Site-to-Site VPN routing between edge firewalls.

4. S/MIME Certificate Management in Large Organization

Problem Context & Given Parameters

- Enterprise Scale: 12,000 Employees
- X.509 S/MIME Certificate Size: 2 KB (2,048 Bytes)
- Renewal Frequency: Annual (Once every year)
- Renewal Overlap: Old and new certificates retained simultaneously for 1 month

Step-by-Step Mathematical Evaluation

A. Total Storage Required During the Renewal Period:

1. Total active certificates per employee during overlap = 2 (1 Old Certificate + 1 New Certificate)
2. Total certificates stored organization-wide = 12,000 employees × 2 = 24,000 certificates
3. Total Certificate Storage = 24,000 × 2 KB = 48,000 KB
= 48.00 MB (or 46.875 MiB)
(Note: Associated private keys, key escrow history, and certificate chains add an estimated 100–150 MB database overhead).

B. Certificate Distribution Overhead Across the Network:

1. Raw distribution payload for annual renewals = 12,000 × 2 KB = 24,000 KB = 24.00 MB
2. Global Address List (GAL) / LDAP Directory Sync Overhead: When 12,000 public certificates are

updated in Active Directory / LDAP, directory synchronization storms across branch DCs generate $12,000 \times 2 \text{ KB} \times N$ domain controllers.

3. Client Revocation Validation Overhead: Continuous OCSP requests and daily Certificate Revocation List (CRL) downloads by thousands of mail clients generate significant recurring network bandwidth.

C. Challenges in Certificate Lifecycle Management at Scale:

- Expiration & Operational Outages: Unmanaged certificate expirations cause immediate email decryption failures and broken digital signature trust across clients.
- Key Escrow & Historical Decryption: While signing keys must never be escrowed (to enforce non-repudiation), encryption private keys MUST be securely archived in an enterprise key escrow system. If a user departs or loses their key, historical encrypted communications become permanently lost.
- CRL Bloat & OCSP Latency: Large enterprises experience multi-megabyte CRL files that choke edge bandwidth and cause email client UI freezes during certificate path validation.
- Mobile & BYOD Provisioning: Distributing identity certs and private keys to iOS, Android, and macOS endpoints securely without user tampering requires complex MDM integration.

D. Proposed Optimization & Modernization Strategies:

- Automated Lifecycle Protocols (SCEP / EST / ACME): Deploy Simple Certificate Enrollment Protocol (SCEP) or Enrollment over Secure Transport (EST) integrated with Microsoft Intune/MDM for zero-touch automatic certificate enrollment and renewal.
- Dual-Key Centralized PKI Architecture: Implement a segregated Public Key Infrastructure where user Signing Keys are generated on tamper-proof smartcards/YubiKeys (no escrow), while Encryption Keys are generated centrally with automated key escrow.
- Short-Lived Certificates & OCSP Stapling: Reduce certificate validity to 90 days with automated rolling renewals, eliminating reliance on massive CRLs and streamlining revocation.

5. Email Encryption Latency in Cloud-Based Systems

Problem Context & Given Parameters

- Symmetric Encryption (AES-256): 0.2 ms per MB of payload
- Asymmetric Key Exchange (RSA-2048): 5.0 ms per email
- Attachment Payload Size: 8 MB per email
- Cloud Email Ingestion Volume: 8,000 emails per hour

Step-by-Step Mathematical Evaluation

A. Total Encryption Time per Email:

1. Symmetric (AES) Data Encryption Time = $8 \text{ MB} \times 0.2 \text{ ms/MB} = 1.6 \text{ ms}$
2. Asymmetric (RSA) Key Exchange Time = 5.0 ms
3. Total Encryption Latency per Email = $1.6 \text{ ms (AES)} + 5.0 \text{ ms (RSA)} = 6.6 \text{ ms}$

B. Total Processing Time per Hour:

1. Total Processing Time = $8,000 \text{ emails/hour} \times 6.6 \text{ ms/email}$
 $= 52,800 \text{ ms/hour}$

2. In Seconds = $52,800 \text{ ms} / 1,000 = 52.80 \text{ Seconds}$ of cumulative CPU processing time per hour
3. Continuous CPU Utilization on a single core = $52.80 \text{ s} / 3,600 \text{ s} = 1.47\%$ (under uniform distribution). However, during burst periods (e.g., 500 emails/min), latency spikes without concurrency.

C. Performance Bottleneck Analysis (Symmetric vs. Asymmetric Operations):

- Latency Breakdown:
 - RSA Key Exchange: $5.0 \text{ ms} / 6.6 \text{ ms} = 75.76\%$ of total compute time
 - AES Data Encryption: $1.6 \text{ ms} / 6.6 \text{ ms} = 24.24\%$ of total compute time
- Root Cause Analysis: Despite the AES cipher processing 8,388,608 Bytes (8 MB) and RSA encrypting only a 32-Byte (256-bit) session key, RSA accounts for over 3/4ths of total execution time. RSA modular exponentiation over large Galois fields (2048-bit integers) is computationally intensive ($O(k^3)$ complexity), whereas AES benefits from hardware pipelining and constant-time S-box operations.

D. Recommended High-Throughput Cloud Optimization Strategies:

- Transition to Elliptic Curve Cryptography (ECDH / X25519): Replacing RSA-2048 with ECDH / X25519 reduces asymmetric key exchange time from 5.0 ms to $< 0.5 \text{ ms}$ (over 10× speedup) while providing equivalent 128-bit cryptographic strength.
- Hardware Acceleration (Intel QAT & AES-NI): Offload asymmetric modular arithmetic to Intel QuickAssist Technology (QAT) accelerators or dedicated cloud crypto co-processors (AWS CloudHSM / Nitro Enclaves).
- Multi-Core Parallel Thread Pools & Asynchronous I/O: Deploy worker pools using Go goroutines or Python AsyncIO + multiprocessing to execute symmetric chunk encryption and asymmetric handshakes across multi-threaded vCPUs in parallel.